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ITEC-617: Information and Technology

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Company: Deere & Company (John Deere)

DT Initiative: From Hardware to SaaS, AI, Autonomy & Precision Agriculture

GitHub: <https://github.com/karakusihsaniye-droid/deere-company-john-deere-dt-case-study>

Introduction

This report summarizes the research, production, and key learnings from a case study based on *Karakus_Ihsaniye_Main_Case Study*, which examines John Deere's digital transformation. The case documents how Deere, an 187-year-old agricultural equipment manufacturer, is reinventing itself as a software and subscription platform using AI and autonomous systems. Research was conducted using Claude Sonnet (claude.ai) across 17 sources, including the Deere 2025 Annual Report (Form 10-K), Reuters, and WIRED.

Key Insights

The protagonist is CEO John C. May, a career Deere insider who in 2020 restructured the company around customer outcomes rather than product lines. The two flagship AI initiatives are See & Spray, reducing herbicide use by 76% on average via computer vision (Iowa State University, 2024), and the 8R autonomous tractor, delivering 15–20% farm productivity gains (Telang, 2025). The case is strategically compelling because three forces collide simultaneously: revenue fell from \$61.3B (FY2023) to \$45.7B (FY2025) while R&D held flat at \$2.31B; the proprietary ecosystem enabling per-acre SaaS pricing triggered a \$99M right-to-repair settlement (Reuters, 2026) and an active FTC antitrust suit; and Choudary (2026) identifies the deepest tension, Deere is shifting decision-making control from farmers to its machines. Whether Deere can build an 85% gross-margin software business while being sued for the ecosystem enabling it is the unresolved core of the case.

Personal Reflection

Working with Claude revealed that AI synthesizes across sources reliably but misplaces facts in the wrong fiscal year. Three concrete errors were caught and corrected: Claude attributed FY2024's \$51.7B revenue to FY2025 (confirmed by 10-K); described six cameras uniformly on

all autonomous tractors (corrected to six-to-sixteen by generation, per TSIA podcast and Deere's January 2025 release); and rounded the Blue River Technology acquisition to \$300M (confirmed as \$305M by WIRED, 2021, and Telang, 2025). Building a chronological anchor table from session one, rather than after the draft, would have caught these earlier. About the transformation itself: I expected a technology story and found a story about control. The right-to-repair litigation is a direct confrontation between Deere's outcome-logic model and a farmer's century-old expectation of owning the tools they buy.

Application

This project used multiple AI tools, including Claude Sonnet (claude.ai), Plexy, and ChatGPT, each serving different roles in the workflow. Claude was primarily used for research synthesis and drafting, while ChatGPT supported structuring and refinement, and Plexy was used for supplementary exploration and cross-checking.

Source-first, section-by-section prompting produced more accurate output than any broad request. Asking Claude to "write the opening scene using only the provided sources, anchored to the November 2025 earnings call" yielded tighter, more verifiable prose than "write the case study." Specifying "cite the exact 10-K line for each figure" eliminated year-mixing errors entirely. In future projects, a dedicated contradiction-checking session, asking Claude to compare its draft against each source before manual review, would be added from the start. The John Deere case sharpens ITEC-617's platform ecosystem framework: Deere replicates Apple's closed-ecosystem model with identical margin logic and identical regulatory vulnerability.

Conclusion

John Deere's transformation tests whether an industrial incumbent can earn subscription trust from customers it is simultaneously suing over repair rights. AI compressed weeks of research

into days; human verification kept it accurate. Proprietary ecosystems create the highest value and the highest resistance, navigating that tension requires judgment no tool can fully automate.